ELSEVIER

Atherosclerosis 122 (1996) 33–46

atherosclerosis

Incomplete dominance of type III hyperlipoproteinemia is associated with the rare apolipoprotein E2 (Arg136 → Ser) variant in multigenerational pedigree studies

M. Pocovi[a], A. Cenarro[a], F. Civeira[a], R.H. Myers[b], E. Casao[a], M. Esteban[a], J.M. Ordovas[c,*]

[a]*Department of Biochemistry and Molecular and Cellular Biology, University of Zaragoza, Zaragoza, Spain*
[b]*Department of Neurology, Boston University School of Medicine, Boston, MA, USA*
[c]*Lipid Metabolism Laboratory, JM-USDA Human Nutrition Research Center at Tufts University, 711 Washington St, Boston, MA 02111, USA*

Received 10 April 1995; revised 5 September 1995; accepted 19 October 1995

Abstract

In the process of screening apolipoprotein (apo) E genotypes in a population of subjects with lipid abnormalities, we have identified five subjects (one homozygote and four heterozygotes) with an abnormal 109 base pairs band following apo E restriction isotyping of amplified DNA with the restriction endonuclease *Cfo*I. The polymerase chain reaction (PCR) products were cloned and their sequencing revealed a C → A substitution at the first nucleotide of codon 136. This mutation resulted in an amino acid substitution Arg to Ser, previously described as apo E2 Christchurch. Family studies were carried out for four of the probands. In these kindreds, stepwise multiple regression analyses indicated that 78% of the cholesterol variability in men was predicted by body mass index, age and the rare apo E2 (Arg136 → Ser) variant. In women, age and the apo E2 (Arg136 → Ser) variant predicted 54.9% of the variability in cholesterol levels. Linkage analysis suggested that the presence of the apo E2 (Arg136 → Ser) variant was linked with the occurrence of cholesterol enriched triglyceride rich lipoproteins and with an incomplete dominance of type III hyperlipoproteinemia. Our data indicates that this mutation may be a relatively common cause of dyslipidemia in the Spanish population.

Keywords: Apolipoprotein E; Hyperlipoproteinemia; Lipoproteins; Genetics

1. Introduction

Genetic variation at the apolipoprotein E gene locus has been proposed as an important determinant of coronary heart disease (CHD) risk [1,2]. Population studies have shown a significant association between apo E phenotype and plasma total and low-density lipoprotein (LDL) cholesterol levels [3,4]. In addition, some case-control studies have demonstrated an association between apo E genetic variability and CHD risk [5–10].

* Corresponding author. Tel.: +1 617 556 3102; fax: +1 617 556 3103; e-mail: Ordovas_li@HNRC.TUFTS.EDU.

0021-9150/96/$15.00 © 1996 Elsevier Science Ireland Ltd. All rights reserved
SSDI 0021-9150(95)06745-5

Population-based studies have also shown some association [11–14] and in a recent study we have shown that the association of the $\epsilon 4$ allele with CHD remains significant after adjustment by traditional coronary risk factors and lipids [15]. Apo E is carried in plasma by chylomicrons, very-low-density lipoproteins (VLDL) and high-density lipoproteins (HDL) [1]. Apo E serves as the ligand for the LDL receptor and more recently, it has been shown that apo E also acts as one of several ligands for the LDL-receptor-related protein in the liver [1,16,17]; however, the precise contribution in vivo of this catabolic pathway has not been fully assessed. In humans, three common apo E isoforms have been described, designated E2, E3, and E4, and are coded by the $\epsilon 2$, $\epsilon 3$, and $\epsilon 4$ alleles respectively [18]. Apo E2 differs from apo E3 by a single amino-acid substitution at position 158 (Arg → Cys), whereas apo E4 differs from apo E3 by a single amino-acid substitution at position 112 (Cys → Arg) [19]. Apo E3 is the most common isoform found in human populations, followed by apo E4 and apo E2 [1]. In addition, several rare mutations have been described at this gene locus.

When compared with the most common apo E3 isoform, the apo E4 isoform exhibits normal binding characteristics to the LDL receptor, but it has a greater affinity for VLDL and chylomicrons than apo E3, which may result in enhanced liver uptake of these lipoproteins and down regulation of the LDL receptor [2,20]. Conversely, the apo E2 isoform has a significantly reduced affinity for the LDL receptor, this may result in the homozygote state in accumulation of cholesterol-enriched lipoprotein remnants and in some individuals this phenotype is associated with type III hyperlipoproteinemia (HLP) [21]. In addition to this common apo E2 isoform, other rare apo E mutations have been associated with type III HLP: apo E2 (Arg 145 → Cys) [22], apo E2 (Lys146 → Gln) [23], apo E2 Christchurch (Arg136 → Ser) [24], apo E2 (Arg136 → Cys) [25], apo E1 Harrisburg (Lys146 → Glu) [26], apo E1 (Gly127 → Asp, Arg158 → Cys) [27–29], apo E3 (Cys112 → Arg; Arg142 → Cys) [30], apo E3 Leiden [31,32], apo E4 Philadelphia (Glu13 → Lys; Arg 145 → Cys) [33,34] and apo E deficiency [35,36]. Compared with the apo E3 isoform, the in vitro receptor binding activities of these mutants range from 4% in apo E3 (Cys112 → Arg; Arg142 → Cys), to 40% in apo E2 (arg145 → Gln), with apo E3 Leiden having an intermediate activity (25%). Type III HLP is expressed only in 2–10% of those homozygotes for the most common apo E2 (Arg158 → Cys). The mechanisms underlying these discrepancies between the degree of in vitro binding activity and the expression of type III HLP have been recently elucidated [37]. In contrast to the most common apo E2 isoform, some of the above apo E variants have been found invariably linked to the expression of type III HLP, either in the homozygous or heterozygous states, suggesting that some of these mutations are associated with dominant expression of this disorder.

The apo E2 (Arg136 → Ser), originally described as apo E2 Christchurch (E2c), has also been reported to be associated with familial dys-β-lipoproteinemia (FD); however, no homozygotes for this mutation have been previously reported. In the present study, we have identified one homozygote and four heterozygotes among a population of 95 hyperlipidemic subjects. We studied the kindred of four of the probands and found 16 more subjects carrying the E2c variant. This variant was linked to the presence of cholesterol enriched triglyceride rich lipoproteins (TRL) and it may be linked to late penetrance of the type III HLP phenotype.

2. Materials and methods

2.1. Study subjects

A working population of 5567 male subjects, resident in the Spanish province of Zaragoza, was screened during five consecutive years for the presence of cardiovascular risk factors. The age range of these subjects was 30–65 years, and the ethnic background was 100% Caucasian. In this population, hyperlipidemia, defined as previously described [38], was consistently found during each of the five annual screenings in 95 subjects. Apo E genotype and phenotype were carried out in these

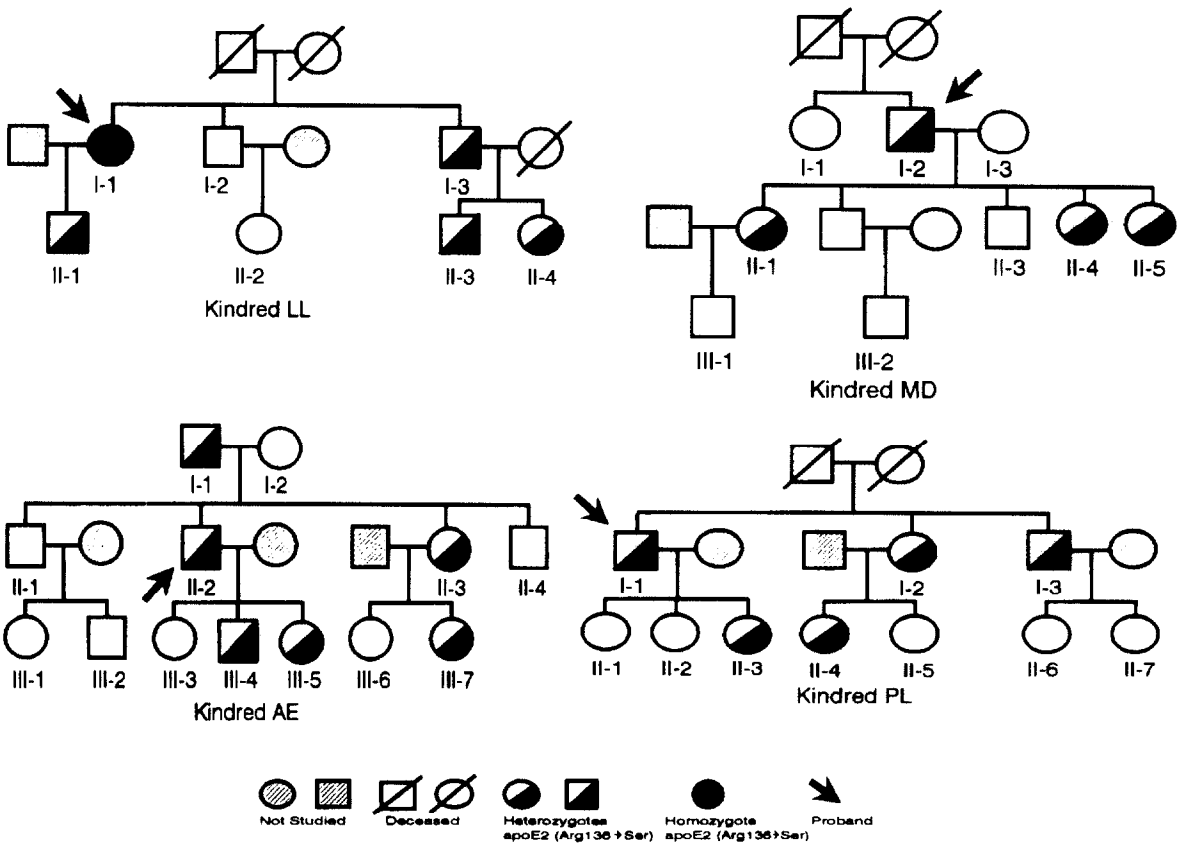


Fig. 1. Pedigrees of the four apoE2 (Arg136→Ser) kindreds. Each of the families is identified by the initials of the proband.

hyperlipidemic subjects. Five subjects were found to be carriers of a rare apo E mutation, apo E2 (Arg136→Ser). Relatives of four of the probands were available for family studies. Pedigrees, including 36 additional subjects are presented in Fig. 1 and Table 1.

2.2. Plasma lipid and lipoprotein determinations

Blood was obtained in EDTA tubes after an overnight (12 h) fast. Plasma was separated by low speed centrifugation (1500 × g), at 4°C for 30 min. Lipoproteins were isolated by sequential ultracentrifugation using solid KBr to adjust the density. Triglycerides and cholesterol in plasma and lipoprotein fractions were determined by enzymatic methods in a laboratory participating in a lipid standardization program. Plasma lipoproteins were visualized after electrophoretic separation on cellulose acetate and Sudan Black staining.

Dys-β-lipoproteinemia was diagnosed by the presence of hypercholesterolemia and hypertriglyceridemia [38], concurrent with a VLDL-C/TG ratio >0.30 and the electrophoresis separation revealed a broad β migrating band.

2.3. Apo E phenotyping and genotyping

Apo E phenotypes were determined by analytical isoelectric focusing (IEF) gel electrophoresis as previously reported [39]. Briefly, VLDL was isolated by ultracentrifugation, desalinated, lyophilized and delipidated with chloroform-methanol (2:1, v/v). The VLDL protein was dissolved in 8 M urea, 10 mM Tris-HCl and 1% dithiothreitol. Electrophocusing was carried out using ampholines in the pH range of 4–6.5.

Table 1
Anthropometric variables, lipid levels, apo E phenotypes and clinical symptoms in kindreds with the apo E2(Arg136→Ser) variant

ID	BMI	Age	Sex	Apo E phenotype	TC	TG	HDLC	VLDLC/TG	Clinical symptoms	Type III HLP
Kindred LL										
I-1	29.8	59	F	E2c/E2c	344	427	33	0.35	Xanthomas	Yes
I-2	32.0	58	M	E3/E3	240	140	47	0.22		
I-3	33.1	47	M	E3/E2c	319	289	35	0.35		Yes
II-1	27.8	38	M	E3/E2c	261	221	36	0.31		Yes
II-2	24.8	32	F	E3/E3	219	91	74	0.08		
II-3	18.5	17	M	E2/E2c	181	79	80	0.29		
II-4	19.7	21	F	E3/E2c	165	79	79	0.31		
Kindred MD										
I-1	23.7	64	F	E3/E3	232	183	79	0.20		
I-2	28.2	56	M	E2/E2c	316	440	32	0.42	Xanthomas, CHD. PVD	Yes
I-3	26.9	50	F	E3/E3	214	61	55	0.16		
II-1	22.0	30	F	E3/E2c	216	74	63	0.34		
II-2	25.3	27	M	E3/E2	157	77	45	0.12		
II-3	22.4	24	M	E3/E2	135	77	51	0.10		
II-4	24.5	22	F	E3/E2c	189	110	40	0.30		
II-5	20.2	17	F	E3/E2c	163	62	52	0.29		
III-1	ND	7	M	E3/E4	218	107	49	0.10		
III-2	ND	3	M	E3/E3	151	61	52	ND		
Kindred AE										
I-1	24.4	65	M	E3/E2c	197	147	26	0.34		
I-2	29.3	61	F	E3/E4	213	83	35	0.19		
II-1	27.6	42	M	E3/E4	221	163	29	0.18		
II-2	29.4	40	M	E4/E2c	302	382	28	0.36	Xanthomas	Yes
II-3	27.6	34	F	E3/E2c	155	48	25	0.31		
II-4	26.4	17	M	E3/E4	157	116	28	0.15		
III-1	22.6	17	F	E3/E3	152	34	37	0.15		
III-2	17.3	13	M	E3/E3	152	32	34	0.18		
III-3	23.6	19	F	E3/E4	206	76	52	0.15		
III-4	23.6	15	M	E3/E2c	143	102	32	0.28		
III-5	20.8	6	F	E3/E2c	173	40	45	0.30		
III-6	18.7	12	F	E3/E3	141	45	29	0.15		
III-7	20.1	9	F	E3/E2c	150	65	27	0.33		
Kindred PL										
I-1	29.8	66	M	E3/E2c	326	401	41	0.32	Stroke	Yes
I-2	28.9	64	F	E3/E2c	274	139	83	0.33		Yes
I-3	30.6	62	M	E3/E2c	277	326	44	0.34		Yes
II-1	22.1	36	F	E3/E2	205	225	39	0.26		
II-2	22.2	34	F	E3/E3	195	110	60	0.14		
II-3	21.9	25	F	E2/E2c	132	138	52	0.31		
II-4	26.0	32	F	E2/E2c	208	176	69	0.49		
II-5	22.2	27	F	E3/E2	229	59	77	0.15		
II-6	25.6	33	F	E3/E3	271	160	59	0.22		
II-7	27.1	29	F	E3/E3	185	105	65	0.10		
Proband DS										
DS	24.0	51	F	E2/E2c	292	180	64	0.42		Yes

BMI: body mass index (kg/m^2); E2c: E2 (Arg136→Ser); TC: plasma cholesterol (mg/dl); TG: plasma triglycerides (mg/dL); HDL-C: cholesterol in HDL (mg/dl); VLDL-C: very low density lipoprotein cholesterol (mg/dl).

Proteins were visualized by staining with Coomassie Blue R-250.

Apo E genotypes were carried out by the polymerase chain reaction (PCR) followed by restriction endonuclease analysis. Genomic DNA was prepared from the total blood cell pellet and the DNA was amplified by PCR in a DNA Thermal Cycler (Perkin Elmer Cetus) using the oligonucleotide primers (F4 and F6) previously described by Hixson and Vernier [40]. The reaction mixture included 0.4 μM of each primer, 250 ng of genomic DNA, 200 μM of each dNTP, 1.5 mM MgCl2, 50 mM KCl and 10 mM Tris-HCl, pH 8.3 in a total volume of 100 μl. Each reaction mixture was heated to 97°C for 30 min before the addition of polymerase. Forty cycles of amplification were carried out using the following conditions: Primer annealing (60°C, 1 min) extension (70°C, 2 min) and denaturation (95°C, 1 min). Amplified DNA fragments were digested with an excess of *Cfo*I (Boehringer Mannheim) for 3 h at 37°C. The digested material was loaded onto a 6% agarose (Nusieve, FMC) gel and electrophoresed for 2 h at 80 V. Bands were visualized using UV illumination (302 nm) after staining the gels with 0.5 μg/ml ethidium bromide for 10 min. Nucleotide substitutions that result in Arg to Cys exchanges at position 112 and 158 alter *Cfo*I cleavage sites, and each genotype can be distinguished by unique combinations of *Cfo*I fragment sizes in all homozygotic and heterozygotic combinations.

Carriers for the apo E2 (Arg136 → Ser) can be identified by an abnormal band of 109 bp, due to the absence of a *Cfo*I cleavage site. In addition, this mutation has been assessed following digestion with *Hph*I. The presence of the Ser136 allele is characterized by the creation of a new restriction site for *Hph*I resulting in two fragments of 152 and 92 bp.

Further characterization of this variant apo E was carried out by cloning and sequencing. Amplified DNA fragments were purified by 2% agarose gel electrophoresis (NuSieve, FMC), digested with the restriction endonucleases EcoRI and *Hind*III and cloned into an *Eco*RI and *Hind*III digested pUC18 vector. DNA sequencing was performed on double stranded DNA using the T7 Sequencing kit (Pharmacia) and M13 sequencing primers.

2.4. Statistical analysis

Results are expressed as mean ± S.D. Statistical significance was determined using ANOVA, followed by the Tukeys HSD test. Stepwise multiple regression analyses were carried out using plasma lipid levels as dependent variables and age, BMI, apo E allele (apo E2, apo E3, apo E4 and apo E2c), and family ID as predictors. Men and women were analyzed separately. Linkage analysis was carried out using the LINKAGE program [41] for both type III HLP and for the VLDL-C/TG > 0.3 phenotypes. An age dependent penetrance curve was applied for type III HLP. Based on the finding of five apo E2c carriers in a population of 5567 subjects, we have estimated the frequency of this mutation in the population as 0.001.

3. Results

3.1. Characterization of the Apo E mutation

In the process of screening apo E genotypes in a population of subjects with lipid abnormalities, we have identified five subjects with an abnormal band of 109 bp following apo E restriction isotyping of amplified DNA with *Cfo*I (Fig. 2). PCR products from each of these subjects were cloned into the pUC18 vector and sequenced. Four independent clones from subject LL were characterized. All of them displayed a C → A substitution at the first nucleotide of codon 136 (Fig. 3). This mutation should result in an amino acid substitution Arg to Ser. Similar analysis was carried out in DNA clones from the other affected subjects (MD, AE, PL, and DS) resulting in the following genotypes: Subjects MD and DS, both had the common apo E2 allele and the apo E2c variant. Subject AE displayed the common apo E4 allele and the apo E2c variant allele, whereas subject PL presented the common apo E3 besides the apo E2c variant.

Further characterization of this variant in these subjects was carried out by endonuclease digestion using *Hph*I (Fig. 4). The homozygous state (LL) was identified by the presence of bands at

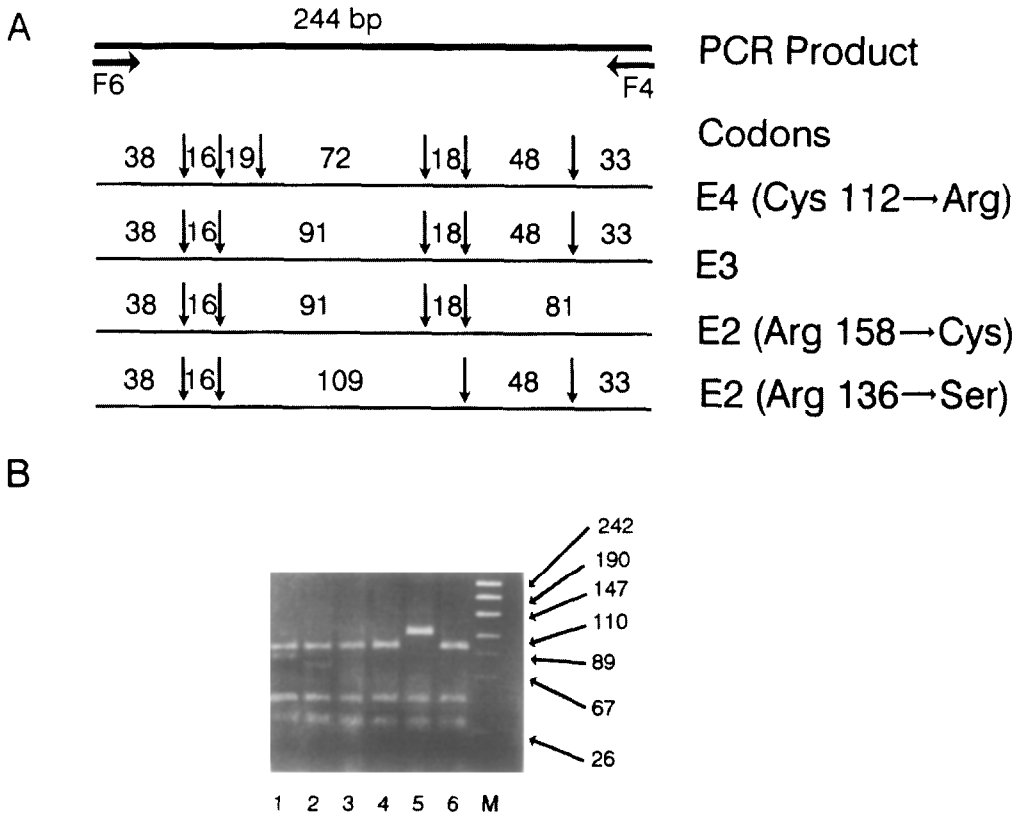


Fig. 2. (A) Schematic representation of the apoE gene exon 4 amplification and the distances (bp) between the CfoI cutting sites that distinguish common apoE isoforms and the variant apoE2(Arg136→Ser). (B) Agarose gel electrophoretic patterns of DNA fragments generated after CfoI digestion are shown for apoE2/3 (lane 1); apoE3/4 (lane 2); apoE3/3 (lanes 3, 4 and 6) and apoE2c/apoE2c (lane 5) subjects. DNA size markers (MspI-digested pUC18, lane M) are shown to the right of the gel.

152 and 92 bp, whereas the heterozygous state (MD, AE, PL and DS) revealed an additional band at 244 bp. This genetic variant had an electrophoretic mobility similar to the most common apo E2 isoform by isoelectric focusing (Fig. 5).

3.2. Family studies

Family studies were carried out for four of the probands (LL, MD, AE, and PL). Their pedigrees are shown in Fig. 1. Anthropometric, plasma lipids and clinical data from all subjects studied are presented in Table 1.

In kindred LL, the proband, I-1, homozygote for the variant apo E2 (E2c) allele displayed xanthomas and type III hyperlipidemia, with elevated plasma total cholesterol (TC) and triglycerides (TG). The VLDL-C to TG ratio presented the increased value associated with this clinical phenotype. The presence of a broad β band was identified after lipoprotein separation on cellulose acetate (data not shown). She was diagnosed with type III HLP at the age of 46 years. One of her siblings (I-3) and one of her children (II-1) were identified as having both the apo E2c variant and type III HLP. The offspring of subject I-3 were the youngest heterozygous carriers of the E2c variant, and both had normal lipid levels; however, they presented VLDL-C/TG values typically associated with dys-β-lipoproteinemia. Subjects I-2 and II-2 presented the apo E3/E3 genotype and normal lipid profile.

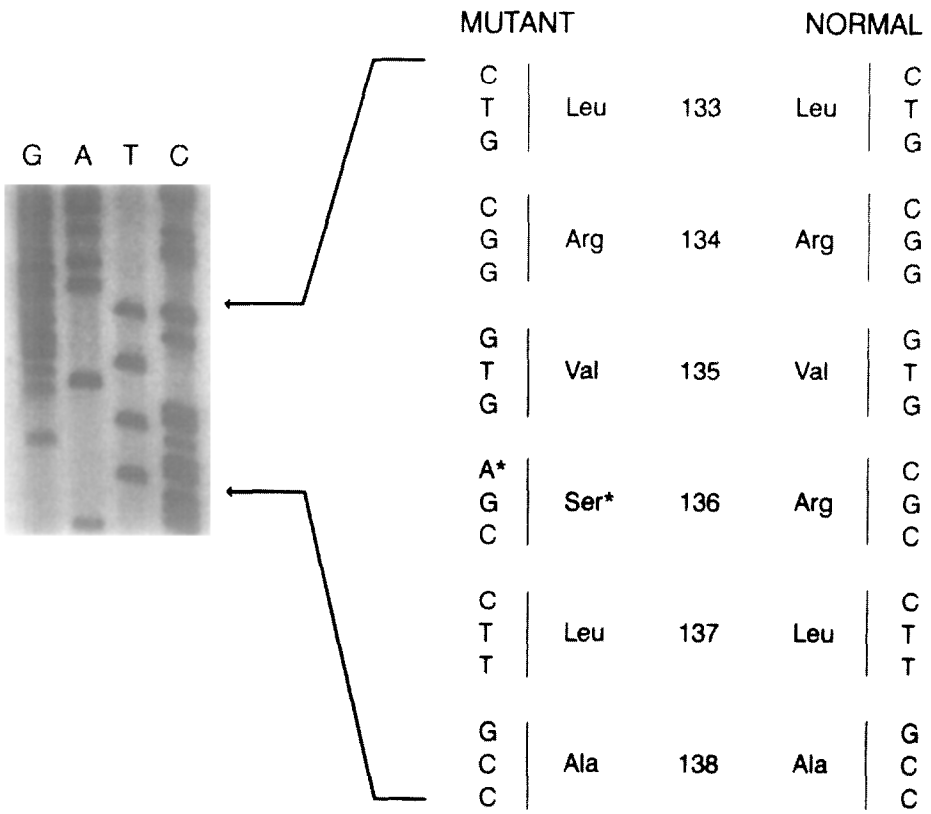


Fig. 3. Sequence of the mutant apoE2 (Arg136→Ser) allele from subject LL I-1. The asterisk denotes the C→A substitution at codon 136.

The proband (I-2) of the kindred MD was a carrier for the common apo E2 and for the apo E2c variant alleles. He displayed the type III HLP phenotype, and he had manifested CHD and peripheral vascular disease (PVD). Three of his children (II-1, II-4 and II-5) were carriers for the apo E2c variant and had normal lipid levels but cholesterol enriched VLDL particles. No other members of this kindred carried the apo E2c variant nor did they present lipid abnormalities.

Three generations were available in kindred AE. The proband, subject II-2 displayed elevated cholesterol and triglyceride levels as well as clinical symptoms of type III HLP, including tuberous xanthomas. In addition, subject I-1, a 65 years-old man, presented elevated VLDL-C/TG ratio, and so did all other carriers of the apo E2c variant.

Kindred PL consisted of three generations. The proband, a 66-year-old male had elevated plasma cholesterol and triglyceride levels, type III HLP and clinical manifestations of stroke. Subject I-2 had both elevated lipid levels, cholesterol enriched VLDL and clinical manifestations of type III HLP. Subjects II-3 and II-4 had dys-β-lipoproteinemia but normal or slightly elevated lipid levels.

Relatives of the fifth proband (DS) did not consent to blood drawing and this kindred was not studied. The proband displayed type III HLP.

Physical characteristics, plasma lipid levels and VLDL-C/TG by gender in carriers and non-carriers of the apo E2c variant are presented in Table 2. Multiple analysis of variance followed by the Tukeys posthoc test was used to compare between groups. BMI and age were included as covariates in the comparison of lipid variables. No signifi-

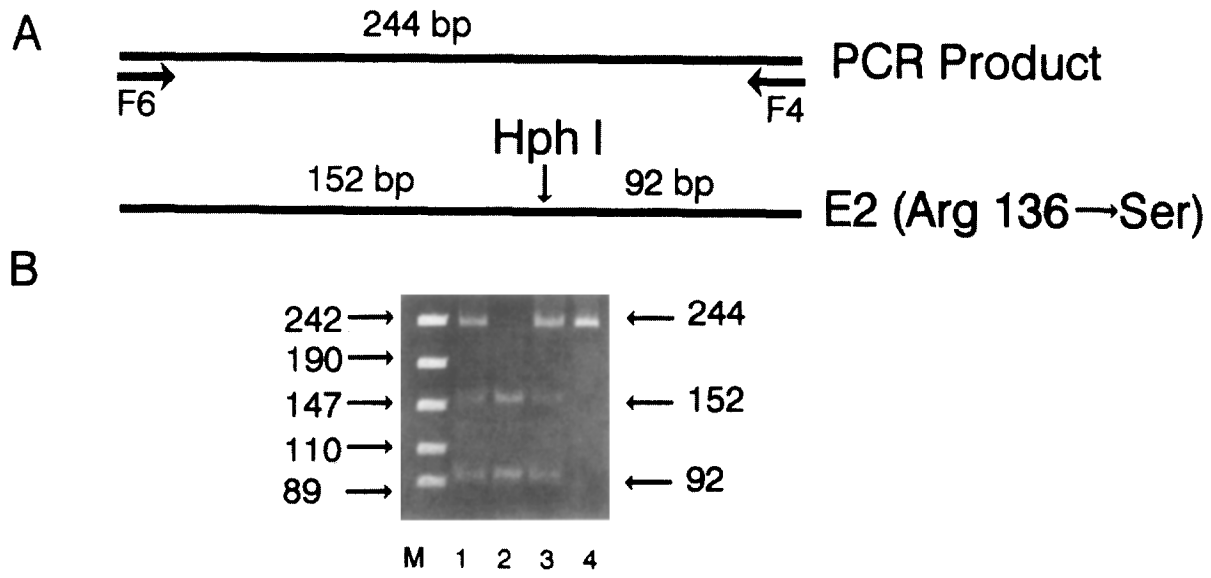


Fig. 4. HphI restriction digest of PCR amplified DNA from three apoE2 (Arg136→Ser) carriers and one control subject. Part of the fourth exon on the apoE gene was amplified as illustrated in the upper panel (A). Restriction digest with enzyme HpHI cleaves the amplified product once in the presence of the apoE2(Arg136→Ser) allele and leads to the production of two fragments of 152 and 92 bp. The lower panel (B) contains the electrophoretic profile of amplified DNA following digestion with the HphI from heterozygotes apoE2(Arg136→Ser), lanes 1 and 3; homozygote apoE2(Arg136→Ser), lane 2 and control subject, lane 4. DNA size markers (MspI-digested pUC18, lane M) are shown to the left of the gel.

cant differences among E2c(−/−) (non-carriers for the apo E2c variant), and E2c carriers (E2c(−/+) and E2c(+/+)) for age and BMI were noted. The heterozygous state for the apo E2c variant was associated with significantly higher plasma cholesterol levels in men, but not in women. The cholesterol level of the only homozygous in this study was significantly higher (344 mg/dl) than those found in female noncarriers (205 ± 35 mg/dl, $P < 0.002$), or heterozygous (192 ± 51 mg/dl, $P < 0.001$) for the apo E2c variant. Similar differences to those described, regarding plasma cholesterol levels, were observed in relation to plasma triglycerides. No significant allele-associated effect was observed regarding HDL cholesterol levels.

The VLDLC to TG ratio was significantly higher in men carrying the apo E2c variant (0.334 ± 0.044) than in non-carriers (0.158 ± 0.044). Similarly, in women, this ratio was significantly higher in carriers of the apo E2c variant (0.339 ± 0.061 and 0.350 in the heterozygous and the homozygous state, respectively) than that noted in non-carriers (0.162 ± 0.049, $P < 0.001$).

Stepwise multiple regression analyses were carried out using plasma lipid levels as dependent variables and age, BMI, apo E allele (apo E2, apo E3, apo E4 and apo E2c variant), and family ID as predictors (Table 3). Men and women were analyzed separately. In men, 78% of the variability in plasma cholesterol levels could be predicted by BMI, apo E2c variant and age, whereas in women, age and apo E2c variant predicted 54.9% of the cholesterol variability. Plasma triglyceride levels were affected by age, apo E2c variant and BMI in men, accounting for 71% of the variability. In women, age and apo E2c variant explained 46.8% of the variability in triglyceride levels. An inverse relationship was noted between BMI and HDL cholesterol levels in men, and between BMI and age in women. However, the percentage of

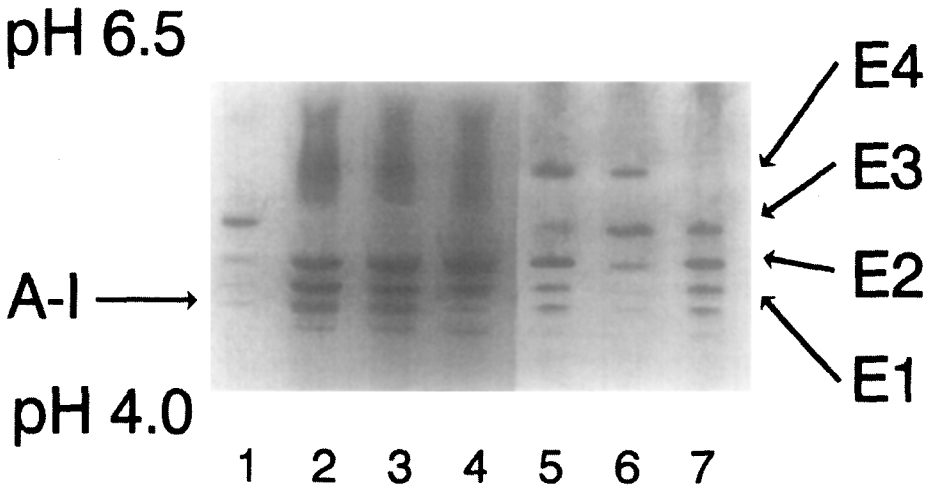


Fig. 5. ApoE isoelectric focusing (pH 4–6.5) patterns of delipidated VLDL: lane 1, subjects with apoE3/3 phenotype; lane 2, apoE2/1 phenotype; lane 3, apoE2/2 phenotype; lane 4 LL I-1, homozygote apoE2(Arg136→Ser); lane 5, subject AE II-2, apoE4/E2(Arg136→Ser); lane 6, AE II-4, apoE3/4 and lane 7, subject AE II-3, apoE3/E2(Arg136→Ser).

the HDL variability explained by this model was only 12% in men and 16% in women. VLDL-C/TG variability was significantly predicted by age, the apo E2c variant and age in men (88%) and by the apo E2c variant in women (67.1%). The more common apo E variants (apo E2, apo E3 and apo E4) did not have a significant effect on the variability of the plasma lipid parameters analyzed in this study. Family ID was not a significant predictor, suggesting that the observed effects associated with the apo E2c variant were kindred independent.

Linkage analysis between type III HLP and the presence of the apo E2c variant was carried out in these kindred. The maximum LOD score for this linkage was 0.83 at $\theta = 0.0$, suggesting that in these pedigrees there was not enough power (too few affected individuals) to demonstrate linkage between the variant allele and type III HLP, probably due to the importance of environmental factors on the expression of this disease. The same analysis was performed for the presence of cholesterol enriched TRL (VLDLC/TG > 0.3). The maximum LOD score was 7.39 at $\theta = 0.0$, suggesting that the presence of the apo E2c variant was linked with the occurrence of cholesterol enriched TRL.

4. Discussion

We describe the presence of a previously reported rare apo E mutation, apo E (Arg136→Ser) [24] in five apparently unrelated Spanish subjects presenting dys-β-lipoproteinemia. One homozygote and four heterozygotes for this variant were identified. Three heterozygotes and the homozygote had elevated cholesterol and triglyceride levels, as well as elevated VLDL-C/TG values consistent with dys-β-lipoproteinemia. Two of them had clinical evidence of vascular disease. One of the heterozygotes had relatively normal lipid values, but still exhibited a high VLDL-C/TG value. Four kindred, including a total of forty subjects, were studied for apo E genotypes by DNA analysis and phenotype by isoelectric focusing, plasma lipid levels, presence of type III hyperlipidemia, and clinical symptoms of vascular disease. Sixteen additional heterozygous subjects for this variant were identified. The presence of the apo E2c variant was always associated with elevated VLDLC/TG in men. Normal lipid levels, in the presence of the variant allele, were observed in younger (mean age, 32 years) and leaner (mean BMI, 22.2) men, whereas the type III HLP phenotype was observed in older (mean age, 51 years) and more obese (mean BMI,

Table 2
Anthropometric and plasma lipid values by apo E Genotype and Gender

Variable	Genotype	MEN	WOMEN
Age (years)	E2c(−/−)	30 ± 17 (6)	35 ± 16 (12)
	E2c(+/−)	45 ± 19 (9)	28 ± 17 (11)
	E2c(+/+)	— (0)	59 (1)
BMI(kg/m^2)	E2c(−/−)	25.2 ± 5.0	24.1 ± 2.9
	E2c(+/−)	27.2 ± 4.4	23.2 ± 3.2
	E2c(+/+)	—	29.8
Plasma cholesterol (mg/100 ml)	E2c(−/−)	177 ± 43	205 ± 35
	E2c(+/−)	258 ± 68*	192 ± 51
	E2c(+/+)	—	344
Plasma triglycerides (mg/100 ml)	E2c(−/−)	101 ± 48	103 ± 59
	E2c(+/−)	262 ± 134*	101 ± 50
	E2c(+/+)	—	427
HDL cholesterol (mg/100 ml)	Ec(−/−)	39 ± 10	55 ± 17
	Ec(+/−)	39 ± 16	54 ± 19
	Ec(+/+)	—	33
VLDLC/TG	E2c(−/−)	0.158 ± 0.044	0.162 ± 0.049
	E2c(+/−)	0.334 ± 0.042*	0.339 ± 0.061*
	E2c(+/+)	—	0.350

Numbers in parentheses represent number of subjects per group.
*$P<0.002$ versus E2c(−/−). MANOVA followed by Tukeys post-hoc test. For lipid variables BMI and age were used as covariates in the analysis.
BMI represents body mass index; VLDL/TG represents very low density lipoprotein to plasma triglycerides ratio. E2c represents the allele apoE(136 Ser).

29.8) male subjects. Similar observations were noted in women. Elevated VLDLC/TG in the presence of normal lipid levels was associated with younger (mean age, 22 years), leaner (mean BMI, 22.5) women as compared to those that presented type III HLP (mean age, 58 years and mean BMI, 28.2). These observations were further demonstrated using stepwise multiple regression analyses. Our results show that presence of the apo E2c variant combined with age and BMI were the major independent predictors of the variability in plasma total cholesterol and triglyceride levels. However, the apo E2c variant was not an independent predictor of HDL-C levels. Our data indicate that the expression of this variant apo E allele is modulated by age and anthropometric characteristics in these subjects. In addition, similar to the findings reported in the original apo E2 Christchurch kindred, we have observed, based on the relative ratio of the different isoforms on IEF (data not shown), that the apo E2c appears to be at a higher concentration than the other apo E allele.

Linkage analysis failed to demonstrate segregation of this rare apo E variant with type III HLP, probably due to both the low number of affected subjects and to the significant influence of environmental factors on the expression of this disease; however, this analysis revealed a significant linkage between this variant and the presence of cholesterol enriched TRL (increased VLDLC/TG ratio).

This mutation has previously been reported in a kindred in New Zealand [24] and in an individual with type III HLP in Utah [42]. No common ancestors could be identified among the kindreds in this study and those previously reported. One hypothesis is that this is the result of a recurrent mutation that has arisen independently in subjects of different genetic background. It is noteworthy to mention that a thymine for cytosine exchange in the first base of the same codon 136, resulting

Table 3
Stepwise multiple regression analyses on plasma lipid values with Age, BMI, apoE allele (E2,E3,E4,E2c), and family ID as predictors

	Dependent variable	Variables in the model	Intercept	β value	B value	Multiple R	Multiple R^2	P value
Men	Plasma	BMI	−42.4	0.49	7.62	0.76	0.58	0.001
	Cholesterol	E2c		0.37	51.28	0.87	0.75	0.014
		Age		0.25	0.92	0.88	0.78	0.259
	Plasma	Age	−263	0.21	1.48	0.69	0.48	0.005
	Triglycerides	E2c		0.44	115.6	0.79	0.62	0.05
		BMI		0.43	12.68	0.84	0.71	0.08
	HDL cholesterol	BMI	66.3	−0.34	−1.03	0.34	0.12	0.21
	VLDLC/TG	E2c	0.118	0.804	0.116	0.910	0.821	0.00001
		Age		0.265	0.001	0.940	0.880	0.029
Women	Plasma	Age	126.8	0.714	2.13	0.718	0.515	0.0001
	Cholesterol	E2c		0.185	16.22	0.751	0.549	0.218
	Plasma	Age	−3.568	0.551	2.70	0.558	0.312	0.004
	Triglycerides	E2c		0.396	59.9	0.684	0.468	0.021
	HDL	Age	88.56	0.594	0.618	0.294	0.90	0.163
	cholesterol	BMI		−0.405	−2.293	0.401	0.161	0.187
	VLDLC/TG	E2c	0.172	0.819	0.146	0.819	0.671	0.00001

Allele types are coded 0 if not present in the pair of alleles, 1 if one is present, and 2 if the specific allele is present in both members of the pair. B, regression coefficient; Ec represents the allele apoE(136 Ser). A different numeric identification was assigned to each kindred, this number was used in the model to ascertain kindred specific effects.

in cysteine for arginine substitution has been reported in two kindred in Canada [25] and Germany [43]. Furthermore, more recently, an Arg136 → His has been reported in Canada, associated with mild dyslipidemia [44]. It is possible that these variants are relatively common, and that, in previous population studies carried out using isoelectric focusing, this variant was classified as the common apo E2 allele. This hypothesis may be supported by the relatively large number of apparently unrelated carriers identified in the present study. These mutations are occurring at the Cytosine of a CpG dinucleotide pair which is the site of frequent point mutations causing human genetic disease [45]. A 14-fold increase in mutations at CpG dinucleotide pairs has been observed compared with that of any one of the remaining 15 dinucleotide pairs.

This mutation is located within the putative receptor binding region and clearly affects the charge of this region [46]. In vitro studies have shown that this mutation results in a 40% affinity of the variant protein for the receptor [47]. The information resulting from these multiple mutations affecting codon 136 underscores the importance of the arginine at this position. In fact, this arginine is conserved in eight of nine species examined so far [48]. Most intriguing is the age and BMI dependent penetration of this allele which supports the concept that expression of type III HLP is conditioned by environmental and other genetic factors. This is clearly shown on subject I-1, from kindred AE. He is a 65-year-old male carrier of the apo E2c variant, with a low BMI. However, although his VLDL-C/TG ratio is > 0.3, this subject did not have type III HLP. Unlike the original proband of the apo E2 Christschurch who also had a low BMI, but carried the most common apo E2 isoform, this subject was apo E2c/E3 and this, in combination with the low BMI, may have protected him from the late onset of the disease

In this work we have described the first homozygote for this apo E2 variant. We have demonstrated that this mutation is relatively common in kindreds from hyperlipidemic subjects in the Spanish population. Furthermore, this mutation is associated with partial dominance of type III HLP.

References

[1] Brown MS, Goldstein JL. A receptor-mediated pathway for cholesterol homeostasis. Science 1986;232:34.
[2] Davignon J, Gregg RE, Sing CF. Apolipoprotein E polymorphism and atherosclerosis. Arteriosclerosis 1988;8:1.
[3] Sing CF, Davignon J. Role of the apolipoprotein E polymorphism in determining normal plasma lipid and lipoprotein variation. Am J Hum Genet 1985;37:268.
[4] Schaefer EJ, Lamon-Fava S, Johnson S et al. Effects of gender and menopausal status on the association of apolipoprotein E phenotype with plasma lipoprotein levels. Arterioscler Thromb 1994;14:1105.
[5] Eto M, Watanabe K, Makino I. Increased frequencies of apolipoprotein E2 and E4 alleles in patients with ischemic heart disease. Clin Genet 1989;13112:253.
[6] Van Bockxmeer FM, Mamotte CDS. Apolipoprotein $\epsilon 4$ homozygosity in young men with coronary heart disease. Lancet 1992;340:879.
[7] Kuusi T, Nieminen MS, Ehnholm C et al. Apolipoprotein E polymorphism and coronary artery disease. Increased prevalence of apolipoprotein E-4 in angiographically verified coronary patients. Arteriosclerosis 1989;9:237.
[8] Nieminen MS, Mattila KJ, Aalto-Setälä K et al. Lipoproteins and their genetic variation in subjects with and without angiographically verified coronary artery disease. Arterioscler Thromb 1992;12:58.
[9] Utermann G, Hardewig A, Zimmer F. Apolipoprotein E phenotypes in patients with myocardial infarction. Hum Genet 1984;65:237.
[10] Stuyt PMJ, Brenninkmeijer BJ, Demacker PNM et al. Apolipoprotein E phenotypes, serum lipoproteins and apolipoproteins in angiographically assessed coronary heart disease. Scand J Clin Lab Invest 1991;51:425.
[11] Menzel HJ, Kladetzky RG, Assmann G. Apolipoprotein E polymorphism and coronary artery disease. Arteriosclerosis 1983;3:310.
[12] Eichner JE, Kuller LH, Orchard TJ et al. Relation of apolipoprotein E phenotype to myocardial infarction and mortality from coronary artery disease. Am J Cardiol 1993;71:160.
[13] Lenzen HJ, Assmann G, Buchwalsky R, Schulte H. Association of Apolipoprotein E polymorphism, low-density lipoprotein cholesterol, and coronary artery disease. Clin Chem 1986;32:778.
[14] Hixson JE, PDAY. Research Group, Apolipoprotein E polymorphisms affect atherosclerosis in young males. Arterioscler Thromb 1991;11:1237.
[15] Wilson PWF, Myers RH, Larson MG, Ordovas JM, Wolf PA, Schaefer EJ. Apolipoprotein E alleles, dyslipidemia, and coronary heart disease: The Framingham Offspring Study. J Am Med Assoc 1994;272:1666.
[16] Choi SY, Cooper AD. A comparison of the roles of the low density lipoprotein (LDL) receptor and the LDL receptor-related protein/α_2-macroglobulin receptor in chylomicron remnant removal in the mouse in vivo. J Biol Chem 1993;268:15804.

[17] Beisiegel U, Weber W, Ihrke G, Herz J, Stanley KK. The LDL-receptor-related protein, LRP, is an apolipoprotein E-binding protein. Nature 1989;341:162.
[18] Zannis VI, Breslow JL. Human very low density lipoprotein apolipoprotein E isoprotein polymorphism is explained by genetic variation and posttranslational modification. Biochemistry 1981;20:1033.
[19] Weisgraber KH, Rall SC Jr., Mahley RW. Human E apoprotein heterogeneity. J Biol Chem 1981;256:9077.
[20] Gregg RE, Zech LA, Schaefer EJ, Stark D, Wilson D, Brewer HB, Jr. Abnormal in vivo metabolism of apolipoprotein E4 in humans. J Clin Invest 1986;78:815.
[21] Utermann G, Hees M, Steinmetz A. Polymorphism of apolipoprotein E and occurrence of dysbetalipoproteinaemia in man. Nature 1977;269:604.
[22] Rall SC Jr., Weisgraber KH, Innerarity TL, Mahley RW. Structural basis for receptor binding heterogeneity of apolipoprotein E from type III hyperlipoproteinemic subjects. Proc Natl Acad Sci USA 1982;79:4696.
[23] Rall SC Jr., Weisgraber KH, Innerarity TL, Bersot TP, Mahley RW. Identification of a new structural variant of human apolipoprotein E, E2(Lys146–Gln), in a type III hyperlipoproteinemic subject with the E3/2 phenotype. J Clin Invest 1983;72:1288.
[24] Wardell MR, Brennan SO, Janus ED, Fraser R, Carrel RW. Apolipoprotein E2-Christchurch (136 Arg–Ser). New variant of human apolipoprotein E in a patient with type III hyperlipoproteinemia. J Clin Invest 1987;80:253.
[25] Walden CC, Huff MW, Leiter LA, Connelly PW, Hegele RA. Detection of a new apolipoprotein-E mutation in type III hyperlipidemia using deoxyribonucleic acid restriction isotyping. J Clin Endocrinol Metab 1994;78:699.
[26] Moriyama K, Sasaki J, Matsunaga A et al. Apolipoprotein E1 Lys-146 → Glu with type III hyperlipoproteinemia. Biochim Biophys Acta Lipids Lipid Metab 1992;1128:58.
[27] Feussner G, Funke H, Weng W, Assmann G, Lackner KJ, Ziegler R. Severe type III hyperlipoproteinemia associated with unusual apolipoprotein E1 phenotype and $\epsilon 1$/'null' genotype. Eur J Clin Invest 1992;22:599.
[28] Feussner G, Ziegler R. Response to therapy of a type III hyperlipoproteinemic subject with the rare apolipoprotein E1 ($Gly_{127} \rightarrow Asp$, $Arg_{158} \rightarrow Cys$) variant. Klin Wochenschr 1992;70:614.
[29] Wenham PR, McDowell IFW, Hodges VM et al. Rare apolipoprotein E variant identified in a patient with Type III hyperlipidaemia. Atherosclerosis 1993;99:261.
[30] Horie Y, Fazio S, Westerlund JR, Weisgraber KH, Rall SC Jr. The functional characteristics of a human apolipoprotein E variant (cysteine at residue 142) may explain its association with dominant expression of type III hyperlipoproteinemia. J Biol Chem 1992;267:1962.
[31] de Knijff P, van den Maagdenberg AMJM, Stalenhoef AFH et al. Familial dysbetalipoproteinemia associated with apolipoprotein E3-Leiden in an extended multigeneration pedigree. J Clin Invest 1991;88:643.
[32] Fazio S, Horie Y, Weisgraber KH, Havekes LM, Rall SC Jr. Preferential association of apolipoprotein E Leiden with very low density lipoproteins of human plasma. J Lipid Res 1993;34:447.
[33] Lohse P, Rader DJ, Brewer HB Jr. Heterozygosity for apolipoprotein $E\text{-}4_{Philadelphia}$ ($Glu^{13} \rightarrow Lys$, $Arg^{145} \rightarrow Cys$) is associated with incomplete dominance of type III hyperlipoproteinemia. J Biol Chem 1992;267:13642.
[34] Lohse P, Mann A, Stein EA, Brewer HB. Apolipoprotein E-4 Philadelphia (Glu13-Lys, Arg145-Cys). J Biol Chem 1991;266:10479.
[35] Schaefer EJ, Gregg RE, Ghiselli G et al. Familial apolipoprotein E deficiency. J Clin Invest 1986;78:1206.
[36] Lohse P, Brewer HB III, Meng MS, Skarlatos SI, Larosa JC, Brewer HB Jr. Familial apolipoprotein E deficiency and type III hyperlipoproteinemia due to a premature stop codon in the apolipoprotein E gene. J Lipid Res 1992;33:1583.
[37] Ji Z-S, Fazio S, Mahley RW. Variable heparan sulfate proteoglycan binding of apolipoprotein E variants may modulate the expression of type III hyperlipoproteinemia. J Biol Chem 1994;269:13421.
[38] Moreda A, Alamillo JA, Pocovi M, Civeira F, Blasco M, Ordovas JM. Lipid levels and lipoprotein distribution in a Spanish working population. Clin Invest Arterioscl 1990;2:48.
[39] Ordovas JM, Litwack-Klein L, Wilson PWF, Schaefer MM, Schaefer, EJ. Apolipoprotein E isoform phenotyping methodology and population frequency with identification of apo E1 and apo E5 isoforms. J Lipid Res 1987;28:371.
[40] Hixson JE, Vernier DT. Restriction isotyping of human apolipoprotein E by gene amplification and cleavage with HhaI. J Lipid Res 1990;31:545.
[41] Lathrop GM, Ott J, Lalouel JM, Julien C. Strategies for multilocus linkage analysis in humans. Proc Natl Acad Sci USA 1984;81:3443.
[42] Emi M, Wu LL, Robertson MA et al. Genotyping and sequence analysis of apolipoprotein E isoforms. Genomics 1988;3:373.
[43] Feussner G, Albanese M, Valencia A, Schuster H. Apolipoprotein E2Heidelberg (arg136 → Cys), a new variant of apolipoprotein E associated with incomplete dominance of type III hyperlipidemia. Atherosclerosis 1994;109:261.
[44] Minnich A, Weisgraber KH, Newhouse Y et al. Identification and characterization of a novel apolipoprotein E variant, apolipoprotein E3' ($Arg_{136} \rightarrow His$): Association with mild dyslipidemia and double pre-β very low density lipoproteins. J Lipid Res 1995;36:57.
[45] Cooper DN, Krawczak M. The mutational spectrum of single base-pair substitutions causing human genetic disease: patterns and predictions. Hum Genet 1990;85:55.
[46] Weisgraber KH, Innerarity TL, Harder KJ et al. The receptor binding domain of human apolipoprotein E. Monoclonal antibody inhibition of binding. J Biol Chem 1983;258:12348.

[47] Lalazar A, Weisgraber KH, Rall SC et al. Site-specific mutagenesis of human apolipoprotein E. Receptor binding activity of variants with single amino acid substitutions. J Biol Chem 1988;263:3542.

[48] Yang Y-W, Chan L, Li W-H. Cloning and sequencing of bovine apolipoprotein E complementary DNA and molecular evolution of apolipoproteins E, C-I, and C-II. J Mol Evol 1991;32:469.